

Governed measurement for room-temperature nitrogen-vacancy quantum registers

An application of the decision-governance overlay, with a prospective validation protocol and a simulated evaluation

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Abstract

Room-temperature nitrogen-vacancy centres in diamond support coherent control of small electron–nuclear spin registers, but measurements acquired during drift or calibration failure can be difficult to distinguish from valid experimental data. This paper proposes an operational quality-control implementation of the Initial Condition Ethics framework, termed ICE, and a prospective protocol for evaluating it. The initial system comprises one NV electron spin and one controllable nuclear spin. A software governor combines time-stamped calibration records, auxiliary measurements and dedicated diagnostic circuits to assign an acceptance decision to each acquisition batch. Separate preflight and post-acquisition records link the intended experiment to the data actually collected. Cryptographic hashes and digital signatures support record integrity and attribution; they do not establish physical truth or computational correctness. The proposed validation compares ICE with ordinary periodic calibration using randomized, blinded perturbations and withheld Bell-state measurements. Primary outcomes are acceptance of reference-classified degraded batches, rejection of reference-classified good batches and useful data yield per unit acquisition time. Statistical uncertainty, indeterminate reference labels and rejected data remain part of the analysis. Published hardware results establish the feasibility of the local register, but no new ICE hardware measurements or detection-performance results are reported here. Multi-register comparison, water-based thermal interfaces and radiation monitoring are identified as separate extensions. The intended contribution is a falsifiable quality-control protocol with auditable experimental records, not a general certification method for arbitrary quantum computations. The governance mechanism is an application of a general decision-governance overlay specified and tested separately [17]. A simulation of the decision logic, calibrated, frozen and tested on held-out sessions, is also reported; it is evidence about the protocol's design, not about device performance.

Keywords: nitrogen-vacancy centre; quantum quality control; calibration drift; experimental provenance; Bell-state fidelity; validation protocol

1 Introduction

Experimental quantum information processing depends on more than the nominal fidelity of a gate. A result also depends on the validity of its state preparation, optical readout, control calibration and analysis during acquisition. Changes in any of these conditions can make a previously characterized device behave differently. Room-temperature NV registers provide a suitable setting in which to study this problem because small, controllable electron–nuclear spin systems have already been demonstrated [1–3].

ICE addresses the operational question of when an acquisition batch should be accepted for a specified experimental purpose. It attaches a versioned decision policy to the evidence available before and during that batch, and preserves the evidence needed to review the decision afterwards. The name Initial Condition Ethics is retained from the framework; the scope of this paper is measurement quality and provenance. Passing its tests does not imply ethical or regulatory certification.

This paper applies the three-gate decision-governance overlay described in a companion paper [17] to quantum measurement. The preflight record corresponds to its pre-decision gate, which commits the acceptance policy before acquisition. Monitoring during acquisition corresponds to its mid-decision gate. The completion record corresponds to its post-decision gate, which checks each batch decision against the unchanged policy and enters it in a hash-chained, signed log. The laboratory keeps its own calibration and control; the overlay adds proof of which rules each batch was accepted under.

The distinction between measurement quality and computational correctness is essential. A stable instrument can implement an incorrect circuit. Two instruments can share a calibration bias. A valid signature can authenticate a record containing an erroneous measurement. Conversely, different samples can be legitimate outputs of the same quantum circuit. A useful governor must therefore state what it tests, what it assumes and what it leaves unresolved.

The research question is whether a specified ICE policy reduces acceptance of degraded data relative to an established calibration procedure, without rejecting an unacceptable fraction of good data or imposing excessive measurement overhead. Time-resolved drift analysis and cross-platform quantum comparisons are established research areas [4,5]. No priority claim is made for those ideas, for randomized benchmarking or for cryptographic logging. The proposed contribution is their operational integration into a small-register acceptance protocol, together with a prospective test that can establish whether the integration has value.

This is a methods proposal rather than a report of a completed device. Numerical hardware results below belong to the cited experiments. Numerical acceptance targets later in the paper are proposed design choices. There are no measured ICE success rates, validated production costs or claims of quantum advantage.

2 Published evidence and its limits

Table 1 summarizes the experimental basis for a local NV prototype. The entries refer to different systems, operations and measurement procedures. They must not be combined into a performance specification for a device that has not been built.

Table 1 Relevant published results

Study	Reported result	Implication for this proposal
Rong et al. [1]	At room temperature, average single-qubit gate fidelity 0.999952 and two-qubit gate fidelity 0.992 using optimized control.	Supports calibrated local electron–nuclear gates, not an end-to-end ICE fidelity claim.
Waldherr et al. [2]	Ambient-condition preparation of three-nuclear-spin entangled states with fidelities above 85%, and three-qubit phase-flip error correction.	Establishes that small local registers support nontrivial quantum protocols.
Minnella et al. [3]	A room-temperature four-qubit entangling gate in 14.8 microseconds; reported parallel-gate fidelity 0.92(4), compared with 0.69(3) for their sequential gate.	Provides a recent control architecture and benchmark, not a universal gate-speed or fidelity limit.
Proctor et al. [4]	Detection and tracking of temporal instability, followed by compensation and experimental verification in a trapped-ion system.	Provides relevant existing methods against which a new governor should be assessed.
Elben et al. [5]	A cross-platform state-comparison protocol using randomized measurements; a 10-qubit experiment–theory demonstration.	Supports statistical comparisons rather than equality of individual samples.

Local spin control and remote optical entanglement are distinct capabilities. Bernien et al. connected NV electron spins separated by three metres using photon-mediated heralding in low-temperature apparatus; the two setups operated at 4 K and 8 K [6]. That result cannot be transferred directly to a room-temperature optical network. The initial ICE prototype requires neither a remote photonic connection nor entanglement between separate processors.

Gate fidelity also differs from state-preparation fidelity and readout accuracy. The four-qubit work explicitly identifies state-preparation and measurement limitations at room temperature [3]. A device evaluation must characterize these contributions separately and explain any correction used in the reported state estimate. The largest number of addressable spins in a paper is not, by itself, evidence of a fully connected or fault-tolerant processor of that size.

3 Minimal experimental system

Local register and control

The initial hardware consists of one optically addressable NV electron spin and one individually controllable nuclear spin in the same diamond. A nearby carbon-13 nucleus is one possible implementation. Selection requires measured coupling parameters, resolvable control operations and a readout procedure suitable for the intended correlations. Host-nitrogen implementations are also possible, but a study must use one explicitly characterized encoding throughout a confirmatory comparison.

The apparatus requires a characterized diamond, optical excitation and collection, single-photon detection, microwave control with calibrated amplitude and phase, suitable nuclear-spin control, a stable bias field and synchronized acquisition. Nuclear control may use radio-frequency excitation or calibrated electron-mediated sequences, depending on the register. Optical initialization does not remove the need to

measure initialization error. The microwave transition frequency is determined in the actual bias field rather than assumed to remain at the zero-field splitting.

An existing research setup is the preferred first implementation. A full procurement specification would have to include optics, pulse generation, amplification, timing, mechanical stability, software integration and operator time. A list of selected components is insufficient to establish the total cost of a reproducible instrument.

Measurements available to the governor

The governor consumes a restricted, documented set of inputs. Table 2 separates auxiliary signals from quantum characterizations. Each input has a timestamp, uncertainty or validated operating interval, calibration identifier, maximum permitted age and explicit missing-data status. Limits are selected during a development phase and then frozen for validation.

Table 2 Proposed minimum monitoring inputs

Input	Acquisition role	Important limitation
Resonance and pulse calibration	Dedicated spectroscopy and pulse tests before or between batches.	A calibration is not a continuous measurement of every gate.
Optical reference signal	Laser-power monitor and interleaved bright and dark references.	Optical changes can mimic changes in spin population.
Readout model	Independently characterized photon-count response and any nuclear-to-electron mapping.	A fixed photon count does not imply a universal readout confidence.
Coherence characterization	Periodic Ramsey or echo measurements with the pulse sequence identified.	An idle coherence time does not determine arbitrary circuit fidelity.
Temperature and field proxies	Time-stamped auxiliary measurements at defined locations.	A sensor reading need not equal the local environment at the NV.
Acquisition integrity	Timing continuity, detector status, circuit identity and data-completeness checks.	Passing checks cannot exclude every hardware or software fault.

Conventional fluorescence acquisition is treated as repeated preparation, operation and measurement, not as repeated nondestructive observation of the same unknown electron-spin state. The count model must match the deployed readout method. If single-shot or nuclear-assisted repetitive readout is used, its mapping errors, disturbance and calibration are included explicitly.

The first implementation uses an ordinary, characterized thermal mount. No particular benefit is assigned to an operating temperature of 37 degrees Celsius. The setpoint is selected to provide stable, measured performance within the apparatus limits. A water film and radiation detector are not part of the minimum acceptance policy.

4 Governance decisions and experimental provenance

Units of acquisition

A shot is one execution of preparation, control and measurement. A batch is a predetermined group of shots and diagnostic observations acquired under one policy version and identified control configuration.

A session is an experimental period bounded by substantial recalibration, restart or a scheduled study boundary. The policy applies to batches because many relevant quantities cannot be estimated accurately from a single shot.

The batch identifier is assigned before acquisition. Its record identifies the circuit, pulse implementation, intended number of shots, calibration version and applicable policy. Quantum fidelity measurements consume experimental resources; they are scheduled explicitly and are never represented as passive measurements of the full unknown computational state.

Preflight decision and final disposition

Preflight checks determine whether the instrument is eligible to begin an acquisition. They use only information already available. Missing essential calibration, stale inputs, invalid configuration or failed instrument checks lead to a recorded hold decision. Each instrument must define how acquisition is interrupted safely and the maximum interval of data that can be affected before an interrupt is effective.

During acquisition, auxiliary signals and designated diagnostic circuits are recorded. At batch closure, the governor produces an accept or reject decision and a reason code. A quality value that cannot be estimated with the specified precision is not silently converted into a pass. Post-acquisition acceptance refers to the recorded conditions and the policy's validated scope, not to a promise that every measurement outcome is correct.

(1)

In equation (1), the decision A is generated from calibration evidence C , auxiliary observations S and designated diagnostic measurements V under the frozen policy parameters θ . The subscript j identifies the batch. Measurements reserved for independent assessment in Section 6 are excluded from these inputs. If the instrument automatically recalibrates or changes pulses, the change closes the current configuration and starts a new identified batch.

Record integrity and authentication

A preflight record contains the intended experiment and its prior evidence. A separate completion record contains actual acquisition times, the number of shots attempted and retained, diagnostics, acceptance reasons and references to all raw data. Rejected and interrupted batches receive records too. A preflight digest cannot attest to events that occur later.

The completion manifest includes cryptographic digests of the data objects, the circuit and pulse configuration, the analysis version, the calibration objects and the preceding finalized manifest. A schema version defines the meaning and units of every field. Canonical serialization, for example the JSON Canonicalization Scheme, prevents an identical record from acquiring different byte representations merely through property ordering [7,8].

(2)

Here M is the manifest payload, JCS its canonical serialization and h its SHA-256 digest. M excludes its own digest and signature; an outer record stores M , h and the signature. Large or binary objects are stored separately and referenced by content digest. The schema fixes number encoding and missing-value handling.

Authentication requires more than a digest. The proposed implementation signs the canonical manifest payload using an established digital-signature implementation, for example Ed25519 [9], and records the verification-key identifier. A retained signature supports attribution to that key; authorization and key custody remain operational responsibilities. A trusted copy of the log head or an independently retained

receipt is needed to make deletion or replacement of the latest records detectable. A local hash chain alone cannot establish completeness against a party able to replace the entire chain.

These mechanisms detect specified forms of record alteration and support reproducible association of results with configurations. They do not prove that a sensor was calibrated, that a timestamp was truthful or that the apparatus ran the intended physics. The threat model assumes honest acquisition software and separately maintained trust in the signing key and measurement chain. Adversarial hardware verification and post-quantum cryptographic security are outside this proposal.

5 Quantum measurements and acceptance statistics

Bell state reference measurement

The principal two-qubit test state is

$$(3)$$

For a density operator ρ on the defined two-qubit subspace, the fidelity to this target can be obtained from three Pauli correlations:

$$(4)$$

XX, YY and ZZ denote products of the corresponding Pauli operators on the two qubits. The identity follows by expanding the projector onto the target state. It tests coherence as well as same-basis population correlations. A mixture of 00 and 11 can reproduce the Z-basis population pattern without possessing the target entanglement.

The estimator uses calibrated measurement rotations and the actual photon-count likelihood or validated outcome-assignment model. Uncertainty from finite sampling, bright and dark references, mapping errors and readout calibration must propagate into the fidelity interval. Leakage outside the encoded subspace is estimated separately. If its survival probability is q , the unconditional target-state fidelity is q times equation (4) evaluated on the normalized in-subspace state. The primary quality threshold applies to this unconditional quantity. If it cannot be identified, the assessment remains indeterminate rather than silently substituting a postselected fidelity. Photon non-detection alone is not evidence of leakage. Any conditioning on charge-state selection or readout outcomes is reported with its probability and justification.

A lower confidence bound above 0.5 provides evidence of entanglement for the specified state under trusted measurement assumptions. This is not a loophole-free Bell test or device-independent certification. For operational quality control, the protocol uses a separate, more demanding fidelity threshold. Section 6 proposes 0.90 as an initial engineering threshold; it is not a universal boundary for useful computation.

Worked example: why $S > 2$ is not an acceptance rule

A common acceptance rule for entanglement, a CHSH value S above 2, fails when detection is imperfect. In an accompanying simulation, a purely local classical model with setting-dependent detection reaches $S = 2.69$ at conditional detection efficiency 0.853. It passes the naive rule and stays below the quantum bound of $2\sqrt{2}$. It is rejected only by the detection-aware bound, under which a coincidence-postselected S is evidence against local models only if it exceeds $4/\eta - 2$ [14,15]. Loophole-free experiments closed this gap, including with NV centres in diamond [16]. The governance point is procedural: the acceptance rule, including its efficiency correction, is committed before acquisition and evaluated against the measured η , not against a headline threshold. The simulation code is released with [17].

Gate characterization

Randomized benchmarking can characterize average errors in the implemented gate set [10]. In the simple exponential model,

$$(5)$$

$P(m)$ is the observed survival probability after m Clifford operations, a and b absorb preparation and measurement contributions, p is the fitted decay parameter and d is the dimension of the benchmarked subspace. For two qubits d equals 4. The inferred quantity r with subscript C is an average error per Clifford under the model assumptions, not automatically an error per primitive physical pulse. Compilation, leakage, gate-dependent errors and temporal variation must be considered.

Interleaved randomized benchmarking can estimate the error of a selected gate, subject to its assumptions and bounds [11]. The implemented sequences, their physical gate decomposition, fit diagnostics and confidence intervals are therefore retained. A fitted average error does not prove that a particular output is correct. It is also insufficient to check whether an operation is unitary: the Pauli X operation is unitary even when the intended operation was the identity.

Avoiding selection bias

The acceptance policy must not use the desired answer or the same noisy fidelity estimate later used to claim its success. Dedicated governor inputs and withheld assessment measurements serve different roles. The assessment is still model-dependent, and shared measurement bias is a limitation even when the data partitions are separate.

Accepting only apparently favourable data changes the reported population. Every outcome is therefore accompanied by attempted and accepted batch counts, the fraction rejected, the fraction with indeterminate assessment and acquisition time. Performance among accepted batches is reported alongside performance over all scheduled batches. Good conditional fidelity is not sufficient evidence of an efficient experimental procedure.

Measurement standard

Each quantity in this protocol has one named definition. Bell-state fidelity uses equation (4) with calibrated measurement rotations. Gate error uses the randomized-benchmarking relation of equation (5) [10]. Entanglement acceptance based on CHSH uses the detection-aware bound, not $S > 2$ [14,15]. Simulated pulse fidelity uses the standard Rabi formula for a square pulse on a two-level system [18]. Proportions are reported with 95% intervals, and clustered data require cluster-aware methods. Every reported number is produced by released code, so repeating an analysis gives identical results.

An analysis implementation is used only after it reproduces known values: fidelity 1 for the ideal Bell state and 0.5 for the equal mixture of 00 and 11 (Appendix B); fidelity 1 for an ideal resonant π -pulse; 0.3166, that is $\frac{1}{2} \sin^2(\pi/\sqrt{2})$, at detuning equal to the Rabi frequency; zero for a 2π pulse; and the small-error limit $1 - F \approx (\pi\epsilon/2)^2$. A calculation that fails a known-value check is not used.

6 Prospective validation protocol

Objective and study phases

The primary objective is to estimate the difference in acceptance of reference-classified degraded batches between ICE and an ordinary periodic-calibration policy on the same register. The study first evaluates detection in shadow mode, then tests the operational effect of enabling the policy. Fault injection is

restricted to reversible perturbations within equipment and laboratory safety limits. No destructive device modification is required.

Phase A is a software qualification exercise. Public data and processing code from the recent room-temperature experiment provide a candidate test of data ingestion and reproduction of selected published analyses [12,13]. That exercise has not been performed as part of this paper. It cannot validate a live governor where the dataset lacks the required time ordering, telemetry or intervention labels. Separate simulations can test the decision implementation under specified noise models, but simulated faults and results must remain labelled as such.

Phase B establishes operation of one local register and develops the policy. It characterizes optical response, coherence, control errors and Bell-state measurements before adding fault injections. If initialization and readout errors cannot be separately identified, their combined effect is reported rather than asserting an unsupported decomposition. The pilot selects batch duration, reference-measurement allocation, calibration lifetime and safe perturbation ranges. Pilot data are excluded from the confirmatory comparison.

Phase C evaluates the frozen policy on new sessions. A laboratory running this evaluation would, before collecting data, commit a time-stamped analysis plan specifying the circuit family, fidelity threshold, stopping rule, sample sizes, independent experimental units, exclusions, randomization scheme and inference method. That commitment is itself a pre-decision gate in the sense of [17]. Any policy change during confirmation ends that evaluation and requires a new identified analysis period.

Baseline and comparison conditions

The baseline is the laboratory's documented periodic-calibration procedure, including its existing validity checks and rejection rules. The calibration interval and evidence available to it are fixed during the pilot. It is not an artificially uncalibrated instrument. ICE adds the specified time-resolved inputs and diagnostic schedule. A record-only condition uses the baseline policy with the same provenance format as ICE, allowing the value of logging to be distinguished from the value of acceptance decisions.

In shadow mode, all scheduled batches are acquired and both policies generate decisions from their permitted inputs. Neither policy can access the withheld assessment results or the injection labels before its decisions are finalized. The common data permit paired comparisons of detection decisions. They do not measure the throughput of a system that actively interrupts and recalibrates; that is the purpose of the subsequent enabled-policy experiment.

The enabled-policy experiment randomizes baseline and ICE acquisition windows within sessions. Reset procedures, control configurations and planned fault exposure are matched, and any recalibration cost is included in elapsed time. Randomized ordering and recovery intervals reduce carry-over effects. The resulting comparison assesses both the benefit and the cost of actual holds, aborts and recovery actions.

Perturbations and controls

Fault schedules are generated independently of the governor and concealed until the analysis records are finalized. Conditions include normal operation, single-parameter changes and a prespecified subset of combined changes. Table 3 defines the perturbation families. Their physical magnitudes are selected from pilot measurements and expressed relative to the device's measured linewidth, pulse calibration or operating range, not copied uncritically from another apparatus.

Table 3 Blinded perturbation families

Perturbation	Implementation principle	Assessment requirement
Frequency detuning	Shift the drive relative to the measured transition.	Include small and intermediate offsets and larger offsets that degrade performance without damaging hardware.
Pulse error	Apply a controlled amplitude, duration or phase offset.	Measure actual output degradation; an injected offset is not automatically a bad batch.
Optical change	Change excitation power or collection efficiency within safe limits.	Distinguish spin-state degradation from a change in readout response.
Slow environmental drift	Apply a controlled temperature or field change, or observe independently recorded natural drift.	Record settling, sensor latency and local measurement uncertainty.
Acquisition fault	Inject missing timestamps, stale calibration identifiers or interrupted data transfer in a test environment.	Verify detection without silently discarding incomplete records.
Combined disturbance	Apply predefined combinations at held-out magnitudes.	Test whether performance persists beyond faults used to tune the policy.

Normal-operation blocks are essential for estimating false alarms. Injection times and levels are not supplied as governor features. Repeated shots within one disturbance interval are not independent realizations of the fault; analysis must preserve the clustering created by sessions and perturbation windows.

Withheld assessment and reference labels

Within each batch, Bell-state assessment preparations are distributed across the acquisition interval with randomized measurement-basis order. They are distinct from governor diagnostics but experience the same specified operating condition. The final assessment must evaluate whether the condition was sufficiently stable for a batch-level description. A fast transient that occurs between reference measurements is an explicit detection limit.

A separately executed analysis estimates the Bell-state fidelity and its 95% uncertainty interval after governor decisions are locked. Readout-calibration uncertainty and changes in optical response are included. The assessment is a reference measurement under stated assumptions, not an infallible label. If an injected optical fault invalidates the reference model and cannot be resolved by independent calibration evidence, the batch remains indeterminate.

For an initial operational threshold of 0.90, let L and U be the lower and upper limits of the assessment interval. The reference labels are

$$(6)$$

All other batches are indeterminate. The proposed threshold must be achievable in stable pilot operation before the confirmatory experiment begins. Changing it after seeing validation results is not allowed. A different application can use a different threshold only through a separately specified protocol.

Label uncertainty is reported explicitly. Results include the number of reference-good, reference-bad and indeterminate batches in every condition, sensitivity to calibration assumptions and worst-case assignments of indeterminate batches. Excluding indeterminate batches from conditional metrics must not make an ambiguous experiment appear conclusive. A joint measurement-error model can supplement the classification analysis when its assumptions are justified.

Outcomes and decision rules

The principal error metric is the bad-batch acceptance fraction, abbreviated BAF. Its denominator is the number of reference-bad batches, not the number of accepted batches. The good-batch rejection fraction, GRF, quantifies the competing loss of usable data.

(7)

(8)

Additional outcomes are the reference-bad fraction among accepted batches, total acceptance fraction, fidelity distribution, detection latency, recovery time and reference-good accepted batches per hour. Standardized batch sizes are required for interpreting the last measure. Timing includes diagnostic shots, initialization, readout, processing, holds and recalibration. Evaluation-only assessment time is identified separately and accounted for consistently in both conditions.

The primary inferential comparison is the paired difference in BAF between ICE and baseline during shadow evaluation. In enabled mode, the prespecified randomized-window analysis estimates operational differences. Confidence intervals must respect dependence within sessions and disturbance windows, using a randomization-based or justified cluster-aware method. The analysis plan identifies one primary comparison and controls multiplicity for confirmatory secondary comparisons. Gate-error estimates and individual shot counts do not substitute for the independent units required by this analysis.

The initial practical target is a point estimate of BAF at most half that of baseline, with a 95% upper confidence bound on the BAF difference below zero. Proposed guardrails are a 95% upper confidence bound on GRF no greater than 0.05 and no more than 25% additional acquisition time under matched conditions. These are design targets rather than forecasts. Useful accepted yield is reported even when the targets are met. If baseline BAF is zero, a relative reduction is undefined; absolute differences, uncertainty and the absence of observed baseline failures are reported instead.

Sample size is determined after the pilot and before confirmation. Power simulations use pilot estimates of baseline degradation prevalence, plausible improvement, readout uncertainty, indeterminate-label frequency and correlation between windows. The confirmatory design should target at least 80% power for the prespecified practically relevant effect at a 5% type-I error level. It must also include enough reference-good units to assess the rejection guardrail. A fixed shot count cannot guarantee these properties. Repeated inspection of ordinary fixed-sample confidence intervals is not a valid optional stopping rule; any sequential design requires its own prespecified, validated analysis.

Reporting and falsification

A complete report includes all scheduled windows, deviations, raw observations, finalized decisions, reference classifications, confidence intervals and time accounting. Results are stratified by fault family, magnitude and session as well as pooled using prespecified weights. The deliberately injected fault mixture is not assumed to represent failure prevalence in ordinary use. A natural-operation follow-up is required before making field-reliability claims.

ICE would fail this initial evaluation if it cannot distinguish degraded from good operation at the prespecified trade-off, if improvements disappear under held-out conditions, or if the measurement overhead removes the practical benefit. A null result remains informative about the selected policy. Neither a positive nor a null result establishes performance for circuit families, hardware configurations or failure modes outside the study.

Simulated evaluation of the decision logic

Before any laboratory study, the decision logic was tested in simulation. The model is a driven two-level system with slow random drift in detuning and drive amplitude, blinded step faults, scheduled recalibration every 25 batches, Poisson photon-count readout (0.12 and 0.084 photons per shot for bright and dark states) and withheld reference measurements labelled with 95% Wilson intervals against an operational π -pulse fidelity threshold of 0.95. Each policy was calibrated on five pilot sessions, frozen with a recorded hash, then tested once on twenty new sessions of 200 batches that included fault sizes absent from calibration. These are results about the decision logic, not measurements of an NV device.

A preliminary sweep reproduced the known dependence of π -pulse fidelity on drive strength under a detuning spread of 0.1 MHz: mean fidelity 0.962 at a Rabi frequency of 0.5 MHz, 0.990 at 1 MHz and 0.9996 at 5 MHz. The model omits heating and pulse distortion, which limit drive strength in practice. The evaluation used 5 MHz, fixed before calibration.

Table 4 Simulated evaluation on held-out sessions

Policy	Gate	Calibration objective	Bad accepted (BAF)	Good rejected (GRF)	Targets
v1	Separate limits on detuning and amplitude	Minimise BAF + GRF	0.012	0.074	GRF not met
v2	Predicted fidelity from diagnostics	Minimise BAF, pilot GRF ≤ 0.05	0.029	0.038	Met
v3, one check	As v2 (new sessions)	As v2	0.028	0.033	Met
v3, two independent checks, both pass	Two diagnostics	As v2	0.020	0.040	Met
v3, two independent checks, averaged	Two diagnostics averaged	As v2	0.022	0.038	Met
v3, two checks, 80% shared error	Two diagnostics, both pass	As v2	0.026	0.040	Met
Periodic calibration only	Checks at calibration only	Laboratory schedule	1.000	0.000	Reference

Version 1 failed its false-rejection target: it was calibrated to the wrong objective and used a box of parameter limits that rejected batches whose fidelity was acceptable. It was not re-tuned. Version 2 corrected both errors as a new frozen policy and, tested on unused sessions, met both targets. In this

model periodic calibration accepted every degraded batch because faults arose between calibrations; that is a property of the model and the least favourable case for the baseline. A second diagnostic check reduced accepted bad batches by roughly a quarter when its errors were independent of the first, and gave little benefit when 80% of the error was shared. Independence must therefore be demonstrated, not assumed. The extra diagnostic time of a second check is not counted, so its effect on useful yield per hour remains open.

7 Extensions requiring separate validation

Comparison between independent registers

After local validation, two or three independent registers can run a common circuit family and compare estimates of chosen observables. The protocol specifies those observables, measurement bases, equivalence margins and joint uncertainty before acquisition. Failure to detect a difference is not proof of equivalence. In a simple implementation, a simultaneous confidence interval for each pairwise difference must lie within a prespecified tolerance, with multiplicity accounted for. More general state comparisons may use established randomized-measurement methods [5].

Exact equality of individual samples is unsuitable for a general sampling task. For independent outputs drawn from the same distribution, the probability of three-way agreement is

$$(9)$$

Here $P(x)$ is the probability of outcome x . For a fair binary outcome, agreement occurs with probability $1/4$ even with perfect processors. For a uniform distribution over n -bit strings,

$$(10)$$

Conditioning on unanimous samples generally changes the output distribution to one proportional to $P(x)$ cubed. Requiring unanimity therefore does not provide a generic fidelity improvement. Deterministic-answer problems require their own task-specific error and decision models.

Agreement of estimated observables remains evidence about consistency, not universal correctness. Shared compilation errors, references, calibration procedures and analysis code can produce correlated bias. Separate chips and unequal temperature readings do not establish statistical independence. Independent implementations, known-answer circuits and deliberately shared-error tests are required to examine these limitations. Arbitrary full-distribution or state comparisons may have prohibitive measurement costs; no fixed fifty-qubit boundary or unrestricted scalable-verification claim is made.

Optional water based thermal interface

A water film is a separate engineering hypothesis. Its relevant question is whether it improves measured stability or usable acquisition yield relative to a specified dry thermal mount. Bulk heat-transfer and dielectric calculations can guide design, but cannot establish the temperature at the qubit, the full thermal path or the microwave field delivered by an assembled antenna.

A subsequent crossover study would compare wet and dry configurations at matched setpoint, heat load and control conditions. It would measure local temperature uncertainty, settling, fluorescence, resonance shifts, coherence, pulse calibration and Bell-state performance. Microwave reflection and delivered control amplitude would be characterized in both configurations. Placing an antenna below the chip does not remove the need to test dielectric loading by adjacent material.

If water purity is an experimental variable, the protocol must specify independent measurements appropriate to the claimed contaminants. Conductivity alone is not evidence for the absence of all organic contamination. Optical thermometry must be calibrated for the actual optical path, emissivity and sensor response. The interface is retained only if a reproducible benefit exceeds its measurement uncertainty and maintenance burden. No coherence enhancement from a special interfacial water structure is assumed.

Radiation monitoring

A radiation detector could provide a timestamped auxiliary channel, but an NV-specific study must first establish a relevant association between detector events and qubit or readout failures. Detector geometry, efficiency, false triggers, blind regions and lost acquisition time are part of that study. Evidence from a different qubit technology does not establish an NV lattice-damage rate. Neither a fixed veto window nor a chip-replacement schedule is justified without measurements of the relevant effects.

8 Limitations and practical interpretation

The protocol depends on the validity of its calibration and assessment models. Separation of governor inputs from assessment data reduces a specific form of selection bias, but does not eliminate common systematic error. Imperfect state preparation, pulse rotations and readout can be difficult to identify separately. Claims must be restricted to what the measured data and stated model can resolve.

Monitoring also has finite bandwidth. The required number of diagnostic preparations, readout repetitions and recovery operations can dominate useful acquisition time. A policy that recognizes slow drift may miss short disturbances or faults that do not affect its chosen diagnostic circuits. Batch duration and calibration expiry are therefore experimental parameters, not universal constants.

Provenance is valuable only if records can be independently interpreted and checked. A digest without accessible data, schema definitions and calibration context offers little scientific reproducibility. A signature establishes a relationship to a key rather than to physical reality. None of the proposed logging mechanisms substitutes for measurement validation, error correction or an algorithm-specific verification argument.

The initial application is research quality control on a small, classically checkable register. It does not demonstrate computational advantage, pharmaceutical accuracy, certified cryptographic randomness or regulatory acceptance. A laboratory implementation could be useful without providing any of those capabilities. Its value would lie in lower undetected degradation, clearer diagnosis or improved useful data yield relative to existing practice.

No novelty claim is made for NV optical control, local entanglement, randomized benchmarking, drift tracking, cross-platform state comparison, canonical serialization or digital signatures. Any future claim of a distinct contribution should identify the implemented acceptance policy and the measured improvement over an appropriate baseline. The cost and maintainability of that implementation remain empirical questions.

9 Conclusion

ICE can be formulated as a testable quality-control layer for small room-temperature NV registers. Its acceptance decisions concern defined acquisition conditions and measured evidence. Separate preflight and completion records make those decisions reviewable without confusing cryptographic integrity with quantum correctness.

Published experiments support the proposed local hardware scope. The next scientific step is a frozen-policy comparison with ordinary calibration, using concealed perturbations, withheld reference measurements and full accounting of rejection and acquisition time. The outcome may establish a useful operational improvement or show that its costs exceed its benefits. Extending the system to multiple registers, water interfaces or radiation vetoes requires additional evidence. Until that evidence exists, the defensible contribution is the protocol and its falsifiable criteria, not a demonstrated high-fidelity verification system.

Data and code availability

No new experimental dataset or validated ICE implementation is reported in this paper. The externally published data and processing code identified for an initial reproduction exercise are available from the repositories in references [12,13]. Their availability does not mean they have been reanalysed here. A future implementation should archive its policy, analysis code, dependency versions, calibration records, all acquisition dispositions and appropriately shareable raw data with persistent identifiers. Simulation code for the decision-logic evaluation (`nv_governor_sweep.py`) and the detection-bias example (`bell_bias_sim.py`) is released with [17]. Repository: <https://github.com/aithoughtcouncil-hattie/missing-layer>.

Use of generative artificial intelligence

Generative artificial intelligence assisted drafting, organization and editorial revision of this manuscript. No experimental observations or ICE performance measurements were generated or represented as measured results. Responsibility for the submitted manuscript and its scientific claims rests with the author.

Appendix A Minimum completion record

This proposed schema covers the manifest payload and its outer integrity fields. Implementation must fix field names, units, missing-value handling and canonical encodings, and retain raw detector observations separately from processed estimates.

Table A1 Required information in a finalized batch record

Record category	Required content
Identity and schema	Unique batch and session identifiers; schema and policy versions; device and qubit-map identifiers.
Intended operation	Circuit and compiled pulse digests; encoding; nominal shot allocation; measurement bases; preflight-record digest.
Actual acquisition	Start and stop times with timebase information; attempted, completed and retained shots; interrupted intervals.
Calibration and telemetry	Calibration-object digests; timestamps; uncertainty; validity periods; sensor channels; missing-data flags.
Raw and derived data	Location, byte size and digest for each raw object; analysis-code version; estimator and uncertainty method.
Disposition	Accept or reject; reason codes; policy inputs; hold and recovery events; decision-finalization time.
Validation separation	Identification of withheld assessment data; sealed injection-schedule digest; time of unblinding.
Integrity and authentication	Previous finalized manifest digest; current manifest digest; signature and verification-key identifier; retained external receipt when used.

The acquisition system must detect duplicate identifiers, missing sequence numbers and incomplete finalization. Reprocessing and corrections append linked records, preserving the source data and identifying the changed analysis. Rejected data are retained to measure false alarms and selection effects.

Appendix B Analytical checks for an implementation

The following identities provide small, exact tests of an analysis implementation. They are mathematical checks, not experimental results or substitutes for validation data.

- The target Bell state has $XX = 1$, $YY = -1$ and $ZZ = 1$, giving fidelity 1 in equation (4).
- An equal mixture of 00 and 11 has $XX = 0$, $YY = 0$ and $ZZ = 1$, giving fidelity 0.5 rather than 1.
- Three independent fair bits agree with probability 0.25, even with ideal preparation and measurement.
- If one binary outcome has probability 0.6, conditioning on three-way agreement changes its probability to $0.216 / 0.280$, approximately 0.7714.
- The Pauli X matrix satisfies unitarity but does not implement the identity operation. A test of unitarity alone must not be classified as a gate-correctness test.
- BAF and GRF are undefined when their respective denominators are zero. An implementation must not substitute zero in either case.

- An ideal resonant π -pulse gives fidelity 1; detuning equal to the Rabi frequency gives $\frac{1}{2} \sin^2(\pi/\sqrt{2}) \approx 0.3166$; a 2π pulse gives 0.

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